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Portfolio Website

nikhilcad

nikhilcad

EDUCATION

National Institute of Technology Delhi

Bachelor of Technology (CGPA: 8.44)

2020-24

EXPERIENCE

ThoughtSpot | Member of Technical Staff 3 | Python, Golang, Java

Aug '25 - Present | Hyderabad

- Architected a **distributed, multi-tenant provisioning engine** using **Python, FastAPI, SQLAlchemy, Alembic, and Terraform** on **AWS (EC2, S3)** to automate isolated environment deployments across **3 global regions (USA, Europe, Oceania)**, reducing onboarding deployment timeline from weeks to **under 2 hours** for **3k+ enterprise users**.
- Created **SpotterCode**, a custom **MCP server** handling **200 concurrent requests** via an **async, multi-worker** and horizontally scalable architecture with caching across **3 dedicated agents** (embed, docs, API); enabling **5k+ enterprise users** to embed BI websites via natural language, increasing embed adoption by **20%**.
- Engineered integrations for ThoughtSpot web plugins across **Google Slides, Sheets, Microsoft Excel, and PowerPoint**, enabling embedded analytics for **2k+ enterprise users**.
- Collaborated in building a **Salesforce LWC plugin** to embed ThoughtSpot dashboards directly within Salesforce CRM, reaching **300+ enterprise Salesforce admins**, a high-value segment at the intersection of two enterprise platforms.
- Prototyped **Spotbuild**, an AI-powered prompt-to-BI system that auto-generates Liveboard answers and worksheets, pioneering an approach later adopted by the product's native AI features.

ThoughtSpot | Member of Technical Staff 2 | Java, Golang, React, Typescript, GraphQL

Jul '24 - Jul '25 | Hyderabad

- Achieved a **90% reduction in homepage load time** by migrating the internal Cassandra impressions database to a new schema, replacing per-view lookups with a unified table storing total views per user.
- Introduced **org awareness** to the trending table of **Cassandra** database by creating a new table with altered schema, allowing each isolated org in the instance to have separate trending objects.
- Executed team-led **zero downtime data migration** of **50M+** rows across clusters in **Cassandra** database by implementing a background script that safely transferred data and seamlessly switched over to the new schema.
- Redesigned the user onboarding to highlight **AI-powered Spotter**, guiding users through creating connections, importing data, and asking questions via natural language, resulting in improved feature adoption by **30%**.
- Addressed a **security vulnerability** in the share API by implementing a Redis-based rate limiter after penetration testing exposed a missing rate limit, effectively controlling API request rates and mitigating abuse risks.
- Diagnosed and patched** a Mixpanel overtracking issue ahead of a renewal, **reducing test event volume by 98.5% (20M to 300K)** and **saving \$30K**.
- Added frontend support for **monitor alerts in Slack**, enabling users to receive data updates directly in Slack without opening the BI tool, increasing notification click rate by **20%**.

ThoughtSpot | Software Engineering Intern | React, Typescript, Playwright, Jest

Jan '24 - Jun '24 | Hyderabad

- Migrated schema viewer from Angular to React by creating new React components and making GraphQL calls to the backend. Also rectified design issues in the legacy UI during migration.
- Worked on **Filter and Sort features** for worksheet tables and columns along with one team member. Collaborated with the backend team to update the API to include **tags, authors, and database info** for sorting.
- Rolled out Knowledge Cards that show users information about a selected table including database, schema, and author. Made API calls to the backend.
- Resolved **critical customer-reported bugs**, with an average turnaround of 1-2 days to merge changes into master.

ThoughtSpot | Software Engineering Intern | React, Typescript, Jest

Jun '23 - Jul '23 | Hyderabad

- Streamlined **50+ accessibility improvements** across the product's website.
- Worked with **React.js** and **Typescript** to ensure accessibility by **keyboard and screen readers**, especially for blind and near-sighted users.
- Collaborated with the team to ensure that the website complied with **WAI-ARIA and WCAG** international standards, achieving **Level A** accessibility certification for legal compliance.

PROJECT HIGHLIGHTS

NEAT Twin-Stick Shooter Agent [Code]

- Built a **NEAT (NeuroEvolution of Augmenting Topologies)** agent from scratch in **C++**, evolving both neural-network topology and weights without ML libraries or gradient descent to play a twin-stick space shooter.
- Engineered a **41-feature real-time game-state representation** and self-growing neural architecture, improving score **52x (400 → 20,800)** and survival time **18x (15s → 278s)** over 6,000 generations.
- Designed a **parallelized experiment framework** for population-size sweeps and repeated trials across CPU threads, reducing a 4-repeat, 6,000-generation, population-80 sweep to **12 minutes** wall-clock time.
- Implemented **deterministic seed-based reproducibility** across simulation, mutation, crossover, and selection, alongside automated checkpoint/GIF recording, genome visualization, and input-sensitivity analysis across 1,000+ evolved genomes.

Golang News Reader (UI + Terminal) [Code]

- Developed a full-stack RSS reader with a **Go backend**, **Preact frontend**, and an interactive **Terminal User Interface (TUI)**.
- Implemented a **SQLite-based caching system** and a **RESTful API** in Go for efficient feed management and content delivery.
- Supports full article content extraction, TTS, and integrated podcast media enclosure.
- Designed a **responsive UI** (web + terminal) with keyboard/mouse navigation, feed grouping, and persistent read/unread tracking.

Real Time Chatroom Application [Code+Screenshots]

- Built a multi-platform chat system with **Next.js**, **Clerk authentication**, and **AWS S3 storage**.
- Supported **2GB media uploads** and enabled real-time messaging with **Socket.IO** for real-time bidirectional communication.
- Deployed a working demo used by **50+ test users**.

SKILLS

- **Core:** Java, Golang, TypeScript, React, SQL, GraphQL, Python, C++
- **Frameworks & Libraries:** FastAPI, SQLAlchemy, Alembic, Next.js, Node.js, Flask
- **Tools & Cloud:** AWS (EC2, S3, Lambda, CloudWatch), GCP, Terraform, Docker, Firebase, Git, Jest, Playwright

ACHIEVEMENTS

- Solved **600+** Data Structures and Algorithm problems on **Leetcode** [Link], rated **1764** in contests.
- Achieved a highest rating of **1736 (Expert)** in **Codeforces** [Link].
- Published paper **Study and Analysis of Plant Disease Identification Models** for ICIC3S Conference, available on [IEEE Xplore] DOI: 10.1109/ICIC3S61846.2024.10603350.